

Comparative Analysis of Tax-to-GDP Ratio in Rich and Poor Countries

Tushar and Reet Gupta

Institute of Economic Growth

Dr. William Joe

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Abstract

This study examines the link between taxation and economic growth in both poor and rich countries. It uses fixed-effects panel regressions with GDP and GDP per capita as dependent variables. The results show clear differences between the two groups of countries. In poor countries, a higher tax-to-GDP ratio significantly reduces both GDP and GDP per capita. This suggests that a heavy tax burden slows down growth. However, direct taxes positively impact both GDP and per capita income. Indirect taxes do not significantly affect GDP but have a slight positive connection with GDP per capita. In rich countries, the overall tax-to-GDP ratio does not significantly impact GDP, but it has a weakly positive relationship with GDP per capita. Direct taxes have a small positive effect on GDP but not on GDP per capita. Indirect taxes consistently have a negative effect, especially on overall GDP. These findings demonstrate that the effect of taxation on growth depends on the country's level of development and the type of taxes imposed. For policy, poor countries should focus more on direct taxes to boost growth, while rich countries should reduce their reliance on indirect taxes to maintain long-term progress.

Introduction

Taxes are a critical component of maintaining most countries' governments and economies. The tax-to-GDP ratio represents the government's tax revenue as a percentage of GDP. The higher the tax-to-GDP ratio, the better the country's economic situation. It offers a detailed view of a country's tax revenue and taxation practices. share of its output. The ratio tells us how well the country can afford its expenses. Usually, the developed countries have a higher tax-to-GDP ratio than developing countries.

The World Bank Group assigns the world's economies to four income groups based on GNI per capita(in US\$ \$): low(<1,145), lower-middle(1,146-4,515), upper-middle(4,516-14,005), and high(>14,005). The classifications are updated annually on July 1, based on the GNI per capita of the preceding calendar year. GNI measures are expressed in United States dollars using conversion factors derived according to the Atlas Method, which in its current form was introduced in 1989. The World Bank's income classification aims to reflect a country's level of development, drawing on Atlas GNI per capita as a broadly available indicator of economic capacity.

For our research, we have divided countries into two groups based on their GNI per capita: rich and poor countries. Rich countries include 26 nations classified as upper-middle and high-income, while poor countries consist of 19 low- and lower-middle-income nations from 2010 to 2019. When comparing wealthy (developed) and impoverished (developing) countries, notable differences appear in their tax-to-GDP ratios. Developed countries typically have higher ratios due to well-organized tax systems, broader tax bases, and stronger institutions. Conversely, developing countries usually face lower ratios because of informal economies, weak tax enforcement, and limited administrative capacity. This comparative analysis aims to explore the relationship between the tax-to-GDP ratio and economic growth. We seek to determine whether direct or indirect taxes have a greater impact on GDP and its growth rate, which type of tax encourages growth, and which people are willing to pay. Additionally, we examine how the tax-to-GDP ratio has changed over time in rich and poor countries and whether wealthier nations rely more heavily on one type of tax.

Literature Review

The relationship between taxation and economic growth has been extensively studied across various countries and economic contexts, revealing various outcomes that depend on income levels, tax structures, and policy environments.

Dahal (2020) investigates Nepal's tax-to-GDP ratio and finds that, although the ratio is increasing and approaching levels seen in developed countries, a high tax-to-GDP ratio does not necessarily translate into significant economic growth. Similarly, Çollaku, Balaj, and Hajdini (2021) analyze quarterly data from Kosovo and show that fluctuations in tax revenues harm GDP in the long run, suggesting that frequent tax increases may negatively impact growth.

In contrast, evidence from developed economies shows varied effects. For instance, Stoilova (2013) examines tax burden trends in the 27 EU member states (1995–2010) and concludes that direct taxes strongly support economic growth, whereas indirect taxes are less effective for generating revenue. Shaqiri, Elshani, and Ahmeti (2024) also find that a 1% increase in corporate income tax (CIT) boosts GDP growth by 0.15%, but a 1% rise in VAT share reduces GDP by 0.33%. Furceri and Karras (2008) provide further evidence that a 1% rise in the total tax-to-GDP ratio reduces real GDP per capita by between 0.5% and 1% in the long term.

Studies from developing economies also reveal mixed outcomes. Abd Hakim et al. (2022) explore 47 developed and 90 developing countries and find that direct and indirect taxes negatively affect GDP per capita in developing nations, while direct taxes have a positive impact in developed countries. Bazgan (2018) focuses on Romania (2009–2017) and shows that increases in indirect taxes significantly stimulate growth, while direct taxes play a smaller role. Rehman, Khan, and Tariq (2020) examine 12 Asian countries and conclude that indirect taxes positively influence growth, although the effect differs across countries due to varying economic conditions. Ilaboya and Mgbame (2012), however, report that in Nigeria, indirect taxes have a negative and insignificant relationship with growth, while overall tax revenue strongly supports economic expansion.

African studies reinforce the importance of tax revenue. Babatunde, Ibukun, and Oyeyemi (2017) find a significant positive relationship between tax revenue and GDP in Africa (2004–2013), confirming that taxation can drive economic growth when well managed. Similarly, Ige (2017) shows that in the United States (1986–2015), taxes on goods and services are positively linked with GDP per capita, with a one-unit increase associated with a 1.43 increase in GDP per capita, highlighting the role of tax revenues in funding growth-enhancing projects.

More recent studies have attempted to determine optimal tax burdens. Menescal and Alves (2024), analyzing 41 developing countries (1990–2019), estimate that the optimal tax-to-GDP ratio for low-income countries is around 25.1%, suggesting that many such countries still need to increase tax revenue mobilization by about six percentage points.

In the Indian context, Kumar (2024) finds that while both direct and indirect taxes show positive associations with GDP, their individual contributions are not statistically significant. However, total tax revenue demonstrates a strong positive relationship with GDP, underlining its role as a reliable predictor of economic growth.

Overall, the literature highlights significant variations in the impact of direct and indirect taxation across regions and income groups. Developed countries tend to benefit more from direct taxes, while developing countries often rely on indirect taxation with mixed outcomes. These findings suggest that tax strategies must be tailored to a country's income level, institutional capacity, and economic conditions.

Against this background, the present study aims to conduct a comparative analysis of rich and poor countries from 2010 to 2019. Specifically, it seeks to examine how the tax-to-GDP ratio has evolved over time, the relative impact of direct and indirect taxes, and whether developed countries rely more on one type of tax compared to developing countries

Research Methodology

This study employs the panel fixed effects model to study the relationship between GDP per capita and GDP with tax-to-GDP ratio in 26 rich and 19 poor countries from 2010 to 2019. Additionally, we aim to test the impact of both indirect and direct taxes on GDP per capita. Data were sourced from the World Bank database and UNU Wider.

Descriptive Statistics

```
. summarize GDP_PER_CAPITA_ GDP_ TAX_TO_GDP_RATIO_ INDIRECT_TAXES_ DIRECT_TAX_ GDP_GROWTH
```

Variable	Obs	Mean	Std. Dev.	Min	Max
GDP_PER_CA~	450	22524.31	24934.75	314.5386	103553.8
GDP_	450	1.42e+12	3.15e+12	1.69e+09	2.15e+13
TAX_TO_GDP~	450	15.03709	5.956123	4.098501	28.38631
INDIRECT_T~	448	8.178906	3.121477	.64	16.44
DIRECT_TAX_	445	7.153101	4.840796	.31	25.49
GDP_GROWTH	405	4.207371	9.662201	-32.6112	52.27316

To better understand the effect of the tax-to-GDP ratio on GDP per capita and GDP, we applied a log transformation to both GDP and GDP per capita. This transformation allows us to better assess the impact of a unit change in the tax-to-GDP ratio on GDP and GDP per capita.

Hausman Test

To determine whether to use a fixed-effects or a random-effects model, we conducted a Hausman test. The hypotheses were as follows:

Null Hypothesis (H_0): The Random Effects (RE) estimator is consistent and efficient, indicating that RE is the correct choice.

Alternative Hypothesis (H_1): The RE estimator is inconsistent, indicating that Fixed Effects (FE) is more suitable.

With a p-value of 0.0440, which is less than the 0.05 significance level, we reject the null hypothesis (H_0) and conclude that the fixed-effects model is preferred over the random-effects model.

Error Correction and Multicollinearity Check

The model incorporates robust standard errors to address heteroskedasticity. We also assessed multicollinearity using the Variance Inflation Factor (VIF). An estimated mean VIF value of 1.80 indicates low multicollinearity among the variables, as VIFs below 5 are considered acceptable.

We created a dummy variable to differentiate rich and poor countries and ran the model separately for each group.

Equation (1) represents the panel data multiple regression model where GDP per capita is regressed on the tax-to-GDP ratio, direct taxes, and indirect taxes.

$$GDPPC_{it} = \alpha_i + \beta_1(\text{Tax_to_GDP_Ratio}_{it}) + \beta_2(\text{Direct_Tax}_{it}) + \beta_3(\text{Indirect_Tax}_{it}) + \epsilon_{it} \quad (1)$$

where,

GDPPC = GDP per capita

Tax_to_GDP_Ratio = Tax to GDP Ratio

Direct_Tax = Direct Tax

Indirect_Tax = Indirect Tax

ϵ = Error Term

Equation (2) represents the panel data multiple regression model where GDP per capita is regressed on the tax-to-GDP ratio, direct taxes, and indirect taxes.

$$GDP_{it} = \alpha_i + \beta_1(\text{Tax_to_GDP_Ratio}_{it}) + \beta_2(\text{Direct_Tax}_{it}) + \beta_3(\text{Indirect_Tax}_{it}) + \epsilon_{it} \quad (2)$$

where,

GDP = GDP

Tax_to_GDP_Ratio = Tax to GDP Ratio

Direct_Tax = Direct Tax

Indirect_Tax = Indirect Tax

ϵ = Error Term

Figure 1

Scatter Plot of direct taxes

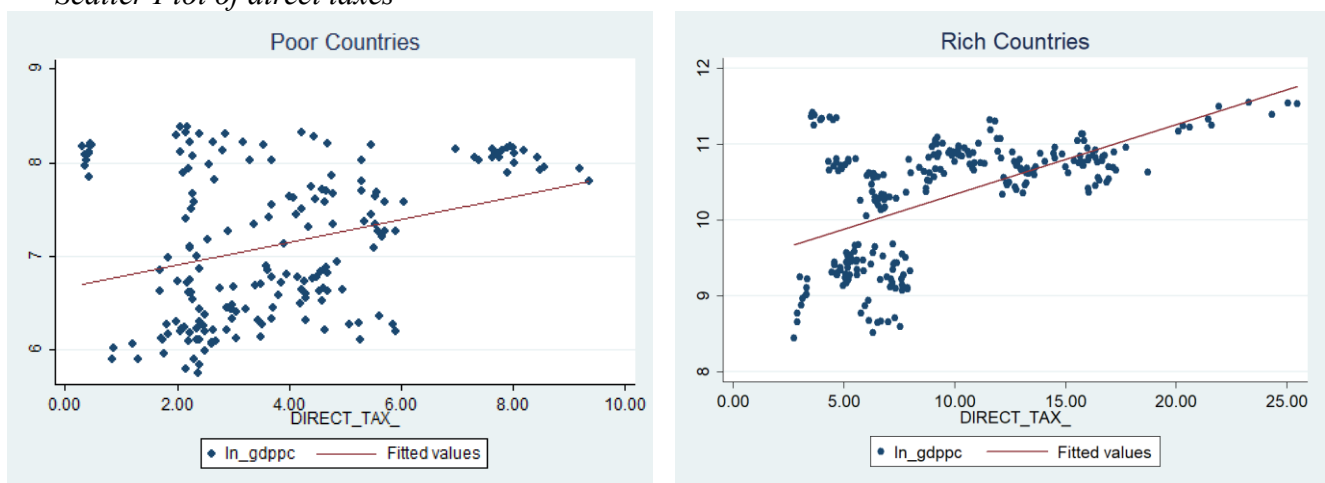


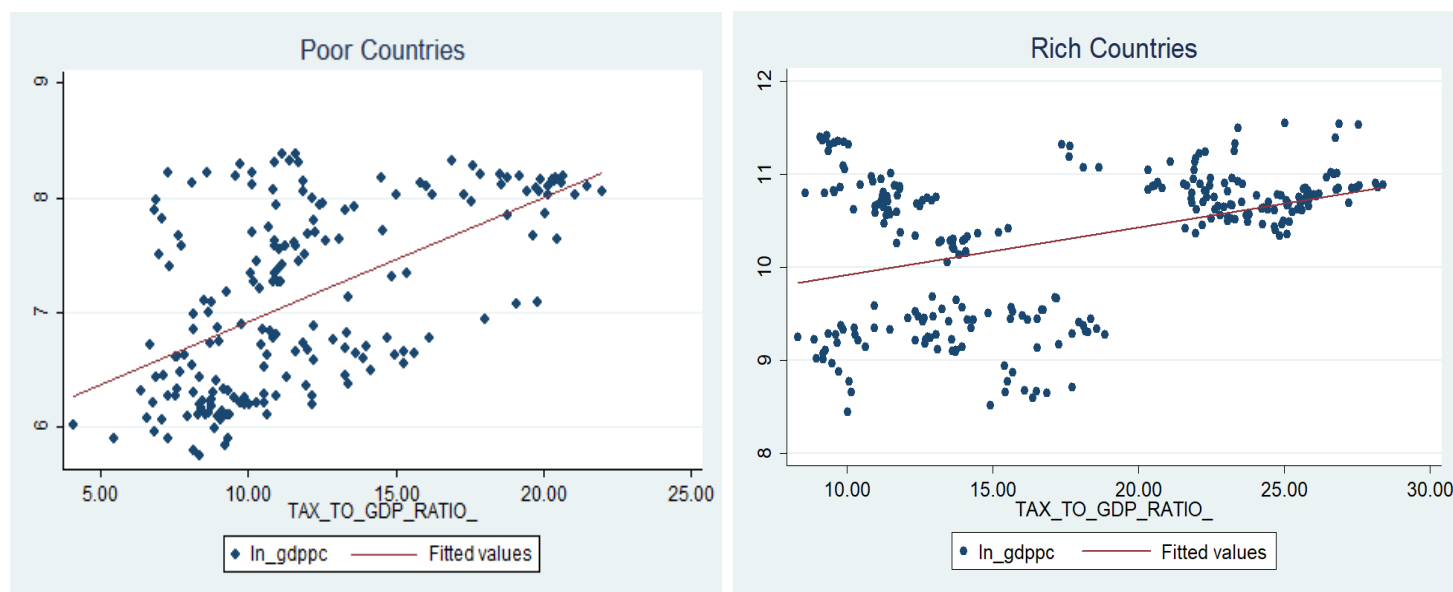
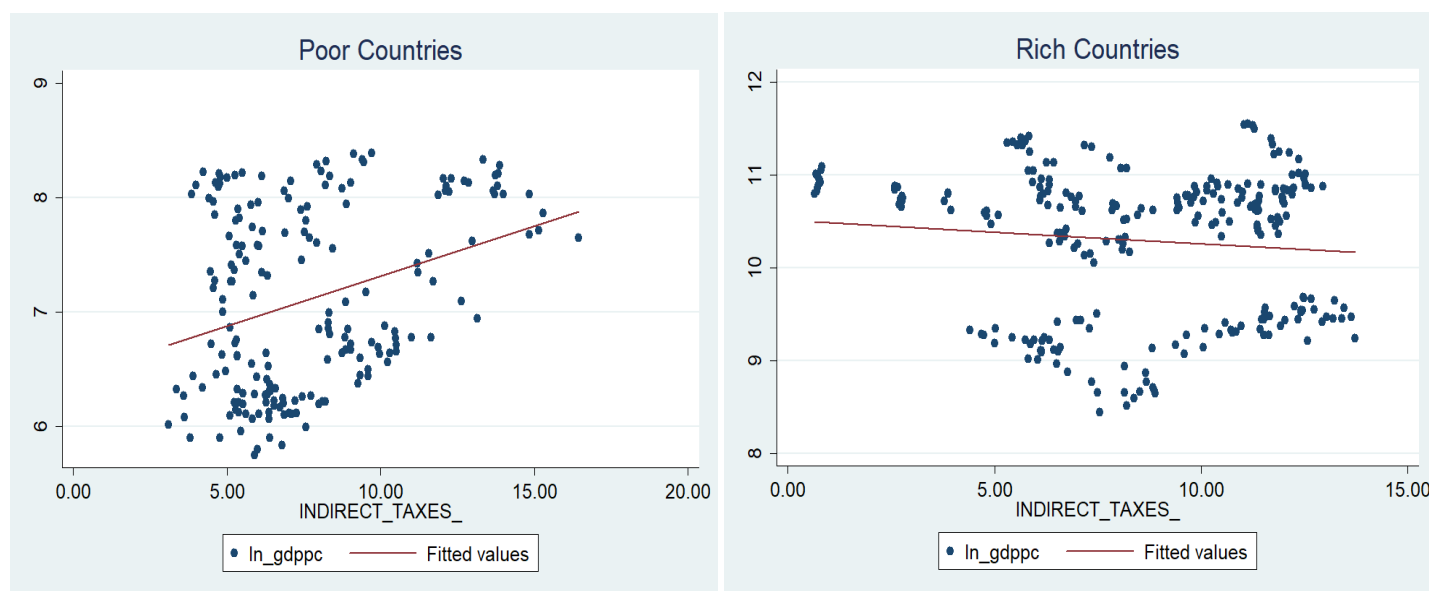
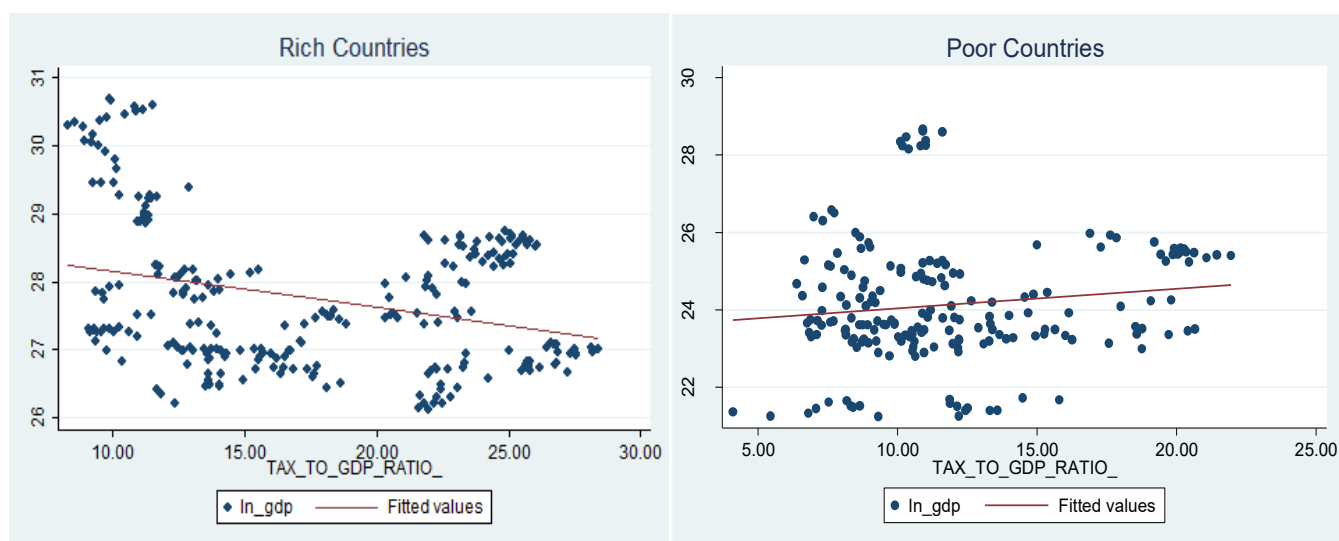
Figure 2*Scatter plot for Tax-to-GDP***Figure 3***Scatter plot for indirect taxes*

Figure 4

Scatter Plot for Tax-to-GDP ratio

**Figure 5**

Scatter Plot for Direct Tax

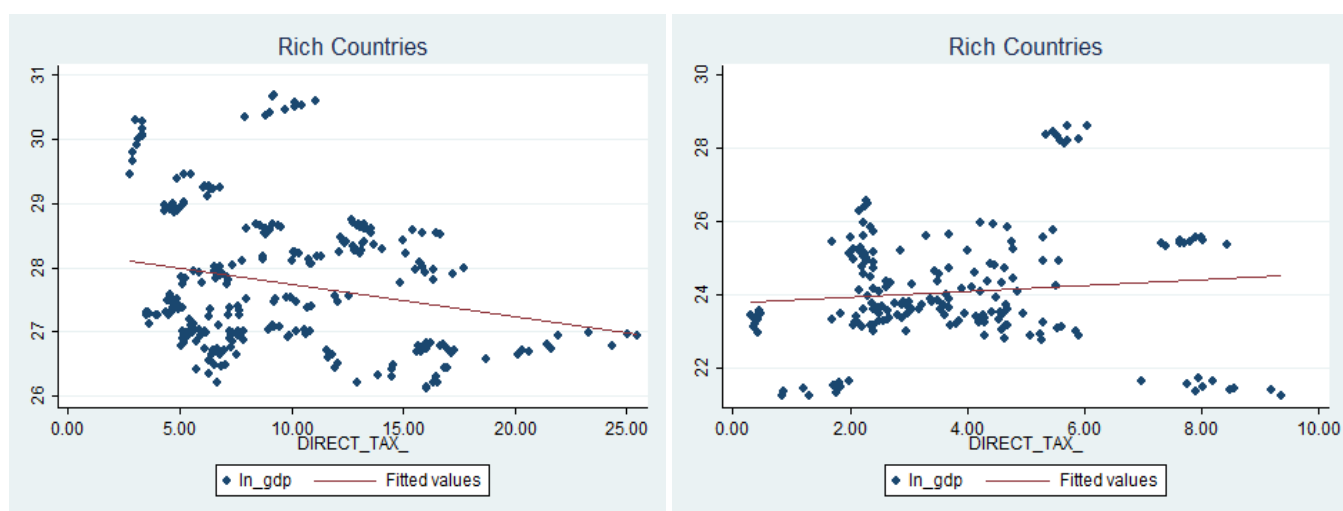
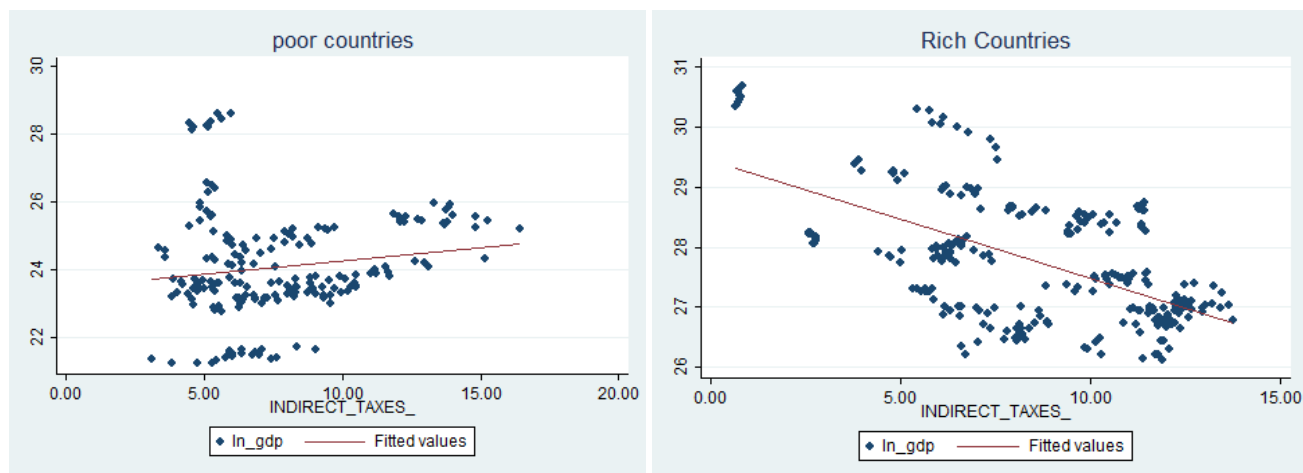


Figure 6

Scatter Plot for Indirect Tax



Results

Fixed Effects Regression for Rich Countries (GDP per capita)

The R-squared value was reported at 0.0409, indicating that the model explains very little variation in GDP per capita among rich countries. The TAX-to-GDP Ratio displayed a marginal significance ($p = 0.076$), suggesting a positive relationship with GDP per capita. However, both direct and indirect taxes exhibited negative coefficients without statistical significance.

Figure 7

Rich Countries Regression Results

Fixed-effects (within) regression		Number of obs	=	1036		
Group variable: country_id		Number of groups	=	26		
R-sq: within	= 0.0426	Obs per group: min	=	36		
between	= 0.0438	avg	=	39.8		
overall	= 0.0409	max	=	40		
corr(u_i, Xb) = 0.0076		F(3,25)	=	1.84		
		Prob > F	=	0.1651		
(Std. Err. adjusted for 26 clusters in country_id)						
ln_gdppc	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
TAX_TO_GDP_RATIO_	.0352661	.0190742	1.85	0.076	-.004018	.0745501
DIRECT_TAX_	-.0270548	.0204691	-1.32	0.198	-.0692118	.0151022
INDIRECT_TAXES_	-.0626318	.0466709	-1.34	0.192	-.1587524	.0334889
_cons	10.47585	.4274661	24.51	0.000	9.595468	11.35623
sigma_u	.76600672					
sigma_e	.10909388					
rho	.98012007	(fraction of variance due to u_i)				

Fixed Effects Regression for Poor Countries (GDP per capita)

In the context of poor countries, the tax-to-GDP ratio was found to have a statistically significant negative effect on GDP per capita at the 10% significance level. The findings also suggest that direct taxes may positively influence growth, albeit this correlation is weaker at the 10% significance level. In contrast, indirect taxes demonstrate a mildly positive yet marginally significant relationship. The overall R-squared value was recorded at 0.0023, but the model fits better within-country, showing a within R-squared of 0.1236, indicating weak explanatory power overall.

Figure 8**Poor Countries Regression Results**

Fixed-effects (within) regression			Number of obs	=	744
Group variable: country_id			Number of groups	=	19
R-sq: within	=	0.1236	Obs per group: min	=	32
between	=	0.0045	avg	=	39.2
overall	=	0.0023	max	=	40
corr(u_i, Xb) = -0.3949			F(3,18)	=	1.97
			Prob > F	=	0.1543
(Std. Err. adjusted for 19 clusters in country_id)					
ln_gdppc	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]
TAX_TO_GDP_RATIO_	-.0656035	.0292124	-2.25	0.038	-.1269765 -.0042306
DIRECT_TAX_	.1297012	.0708487	1.83	0.084	-.0191463 .2785487
INDIRECT_TAXES_	.0575311	.0324189	1.77	0.093	-.0105785 .1256407
_cons	6.970207	.2706346	25.76	0.000	6.401625 7.538789
sigma_u	.85658715				
sigma_e	.17108881				
rho	.96163711	(fraction of variance due to u_i)			

Fixed Effects Regression for Rich Countries (GDP)

The tax-to-GDP ratio shows no significant relationship with GDP per capita, suggesting that the overall level of taxation does not strongly influence growth in high-income nations. However, the composition of taxation appears crucial: direct taxes are positively associated with GDP per capita, significant at the 10% level, reinforcing the argument that direct taxation supports sustainable growth in developed economies. On the other hand, indirect taxes exhibit a negative and significant effect, highlighting their detrimental impact on economic growth in rich countries. This outcome reflects the reliance of developed nations on direct taxation as a more effective and equitable revenue source, while indirect taxes may distort consumption and reduce growth.

Figure 9**Rich Countries Regression Result**

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. xtreg ln_gdp TAX_TO_GDP_RATIO_ DIRECT_TAX_ INDIRECT_TAXES_ if Dummy_Var_2==1,fe robust

```

Fixed-effects (within) regression	Number of obs	=	259
Group variable: country_id	Number of groups	=	26
R-sq: within = 0.1394	Obs per group: min	=	9
between = 0.1309	avg	=	10.0
overall = 0.1296	max	=	10
	F(3,25)	=	3.66
corr(u_i, Xb) = 0.1634	Prob > F	=	0.0259

(Std. Err. adjusted for 26 clusters in country_id)

ln_gdp	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
TAX_TO_GDP_RATIO_	.0079383	.0118833	0.67	0.510	-.0165358	.0324124
DIRECT_TAX_	.0307913	.0175315	1.76	0.091	-.0053155	.0668982
INDIRECT_TAXES_	-.0672261	.0378129	-1.78	0.088	-.1451032	.010651
_cons	27.90455	.3049941	91.49	0.000	27.27641	28.5327
sigma_u	1.0336539					
sigma_e	.11129864					
rho	.98853898	(fraction of variance due to u_i)				

Fixed Effects Regression for Rich Countries (GDP)

The tax-to-GDP ratio is negatively associated with GDP per capita, with the coefficient being marginally significant at the 10% level. This suggests that a higher overall tax burden may hinder economic growth in low-income economies, possibly due to inefficiencies in tax administration and limited capacity to channel tax revenues into productive investment. In contrast, direct taxes display a positive and significant effect on GDP per capita, indicating that revenue from direct taxation supports growth in poor countries. Indirect taxes, although positive, are statistically insignificant, implying that they play a limited role in driving growth in these economies.

Figure 10

Poor Countries Regression Results

```

. xtreg ln_gdp TAX_TO_GDP_RATIO_ DIRECT_TAX_ INDIRECT_TAXES_ if Dummy_Var_2==1,fe robust

```

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TAX_TO_GDP_RATIO_	.0079383	.0118833	0.67	0.510	-.0165358	.0324124
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_cons	27.90455	.3049941	91.49	0.000	27.27641	28.5327
sigma_u	1.0336539					
sigma_e	.11129864					
rho	.98853898	(fraction of variance due to u_i)				

Discussion

The main objective of our research was to explore the relationship between GDP per capita and tax-to-GDP ratio, as well as to assess whether direct or indirect taxes exert a greater influence on economic growth. Using the fixed-effect panel regression model, our findings indicate that the relationship between tax-to-GDP ratio differs across income groups. In rich countries, the tax-to-GDP ratio displayed a positive association with GDP per capita, while in poor countries the association was negative. This divergence may reflect underlying differences in tax systems, tax rates, enforcement mechanisms, levels of tax evasion, and broader socio-economic conditions such as crime, corruption, and institutional quality.

When examining the role of direct and indirect taxes, the coefficients were found to be either statistically insignificant or only marginally significant at the 10% level for both rich and poor countries. This outcome implies that, when countries are analyzed in groups, direct and indirect taxes may not exert a uniform impact on economic growth. The lack of significance could be attributed to variations in tax structures, political stability, inflation, unemployment, and other country-specific indicators.

Taxes are a crucial component of an economy, as they not only provide the revenue necessary to finance government expenditures but also play a role in shaping incentives, investment, and long-term growth. A country's tax-to-GDP ratio serves as an indicator of its fiscal capacity and financial health. However, a high ratio does not automatically guarantee sustained economic growth. For instance, Dahal (2020) found that in Nepal, a higher tax-to-GDP ratio did not translate into significant growth, highlighting the importance of institutional and structural contexts. Similarly, Stoilova (2013), analyzing the 27 EU member states from 1995 to 2010, reported that direct taxes tend to support growth more effectively than indirect taxes, which were less efficient in generating sustainable budget revenue. By contrast, Bazgan (2018), studying Romania from 2009 to 2017, found that indirect taxes significantly boosted growth while direct taxes had a smaller effect.

These contrasting findings emphasize that the impact of taxation on growth is context-dependent. What matters most is not merely the level or type of taxation, but how tax revenues are utilized. Effective allocation of tax revenues toward infrastructure, education, healthcare, and innovation can foster long-term growth, whereas inefficient spending, corruption, and weak institutions can undermine the potential benefits of taxation. Therefore, strong leadership, transparency, and low levels of tax evasion are critical to ensuring that taxation contributes positively to economic performance.

Conclusion

This study sets out to investigate the relationship between GDP per capita and taxation, focusing on the tax-to-GDP ratio and the relative impact on direct and indirect taxes using a fixed-effects panel regression model. Our analysis revealed that the relationship between taxation and economic growth varies significantly across income groups.

For rich countries, the coefficient for the tax-to-GDP ratio was 0.035 and significant at the 10% level, indicating that a one-unit increase in the tax-to-GDP ratio is associated with a 0.035% increase in GDP per capita. However, the coefficients for direct tax (-0.027) and indirect tax (-0.062) were statistically insignificant, suggesting that these tax components do not exert a strong influence on growth when countries are analysed as a group.

For poor countries, the coefficient for the tax-to-GDP ratio was -0.065 and significant at the 10% level, meaning that a one-unit increase in the tax-to-GDP ratio is associated with a 0.065% decline in GDP per capita. The coefficients for direct tax (0.1297) and indirect tax (0.0575) were marginally significant at the 10% level, suggesting a weak positive association but not strong enough to conclude consistent effects.

Taken together, these results imply that countries should not rely solely on the tax-to-GDP ratio as a measure of economic performance. Rich countries appear to convert higher tax revenues into greater economic performance, while poor countries struggle to do so due to weak institutions, governance challenges, and inefficient allocation of resources. This highlights that the effectiveness of taxation in promoting growth depends not only on tax levels but also on how tax revenues are managed.

From a policy perspective, countries should focus on designing tax systems that focus on their political stability, unemployment, inflation, and other macroeconomic indicators. Moreover, improving institutional quality, reducing corruption, and ensuring efficient use of tax revenues in areas such as infrastructure, education, and healthcare are crucial for transforming tax revenues into sustainable economic growth.

References

- Abd Hakim, T., Karia, A. A., David, J., Ginsad, R., Lokman, N., & Zolkafli, S. (2022). Impact of direct and indirect taxes on economic development: A comparison between developed and developing countries. *Cogent Economics & Finance*, 10(1).
<https://doi.org/10.1080/23322039.2022.2141423>
- Babatunde, O. A., Ibukun, A. O., & Oyeyemi, O. G. (2017). Taxation revenue and economic growth in Africa. *Journal of accounting and taxation*, 9(2), 11-22.
- Bazgan, R. M. (2018). The impact of direct and indirect taxes on economic growth: An empirical analysis related to Romania. In *Proceedings of the international conference on business excellence* (Vol. 12, No. 1, pp. 114-127). Sciendo.
- Çollaku, L., Balaj, D., & Hajdini, A. (2023). Correlation between tax revenues and gross domestic product: Evidence from the developing economy. *Corporate and Business Strategy Review*, 4(1), 31-38.
- Dahal, A. K. (2020). Tax-to-GDP Ratio and the Relation of Tax Revenue with GDP: Nepalese Perspective. *Researcher: A Research Journal of Culture and Society*, 4(1), 80–96.
<https://doi.org/10.3126/researcher.v4i1.33813>
- Furceri, D., & Karras, G. (2008). Tax changes and economic growth: Empirical evidence for a panel of OECD countries. *Manuscript*, University of Illinois, 1, 1-29.
- Ige, O. (2017). The Role of Taxation In Driving Economic Growth: Evidence From The United States.
- Ilaboya, O. J., & Mgbame, C. O. (2012). Indirect tax and economic growth. *Research Journal of Finance and Accounting*, 3(11), 70-82.

- Kumar, J. Z. (2024). An empirical analysis on tax-to-GDP ratio in India. *International Journal of Research Publication and Reviews*, 5(2), 3219–3223. <https://www.ijrpr.com>
- Menescal, L., & Alves, J. (2024). Optimal threshold taxation: An empirical investigation for developing economies. *The Journal of Economic Asymmetries*, 29, e00343.
- Rehman, Z. U., Khan, M. A., & Tariq, M. (2020). Indirect taxation and economic growth relationship: empirical evidence from Asian countries. *Pakistan Journal of Humanities & Social Sciences Research*, 3(01), 131-144.
- Shaqiri, V., Elshani, A., & Ahmeti, S. (2024). The Effect of Direct and Indirect Taxes on Economic Growth in Developed Countries. *Ekonomika*, 103(2), 123-139.
- Stoilova, Desislava (2013). An empirical evidence for the impact of taxation on economy growth in the European Union